

GVF Predictive State

Status: independent negative-result proposal and redesign target.

Abstract

This proposal tests whether learned General Value Function predictions can serve as useful agent state under partial observability. In a long aliased T-maze, the agent sees a cue at the start, loses direct access to it in the corridor, and must remember it at the junction. The current main pilot compares raw observation, short history, trace memory, recurrent GVF features, and oracle memory. Trace memory and oracle memory solve the task, while the current recurrent GVF design remains near chance. This is a valuable negative result: it does not refute GVFs, but it shows that the current GVF question/state design fails to carry the hidden cue information required for control.

Standalone Study Summary

This study asks whether GVFs can serve as useful agent state in a partially observable control task. The RL problem is a T-maze where an early cue determines the correct later junction action. The implemented representations are raw observation, short history, hand-coded trace memory, recurrent GVF features, and oracle memory; control uses online linear Sarsa. The experiment measures trial-end accuracy, reward, GVF values, cue traces, GVF TD error, and control TD error. The current evidence is negative: trace and oracle memory solve the task, while the current recurrent GVF representation remains near chance. The result is valuable as a redesign target, not as a claim that GVFs generally fail.

Research Motivation

In partially observable environments, current observation is not state. A long-lived agent must construct state from experience. The Alberta Plan proposes GVFs as a major route: predictions about future signals can become knowledge used by the agent. This idea is powerful, but it has a sharp failure mode. A prediction can be accurate, stable, and still useless for the decision the agent must make.

The goal of this proposal is therefore not to make a decorative auxiliary prediction. The goal is to ask whether a learned prediction can carry missing information that the control policy needs.

Research Question

Can learned GVF predictions supply missing cue information in a partially observable T-maze?

More specifically:

> Does adding a recurrent GVF feature improve trial-end control accuracy over raw observation and short history, and does it approach a cheap trace-memory baseline?

The trace-memory baseline is intentionally strong and cheap. A learned GVF state should not be promoted if a simple non-deep memory trace solves the task and the GVF does not.

Alberta Plan Connection

This proposal targets:

- - GVFs and predictive knowledge;
- - agent-state construction;
- - partial observability;
- - online value learning;
- - useful predictions rather than prediction accuracy alone.

It is highly aligned with the Alberta Plan, but the current result is negative and must be reported honestly.

Related Work

GVF and Horde work motivates learning many predictive questions from experience. Work on useful predictions asks which predictions actually help an agent. Online agent-state and generate-and-test work motivates recurrent trace-like state construction. Recent streaming partial-observability papers show that no-replay, batch-size-one memory remains an active problem, though those papers often use deep recurrent architectures outside this course project's constraints.

Local references:

- - [resources/alberta_plan_related/finding_useful_predictions_2111.11212.pdf](#)
- - [resources/alberta_plan_related/horde_lifelong_offpolicy_1206.6262.pdf](#)
- - [resources/alberta_plan_related/learning_agent_state_online_2112.15236.pdf](#)

External context:

- - Streaming RL under Partial Observability with Real-Time Recurrent Learning: <https://arxiv.org/abs/2605.24709>

Environment

The testbed is a long aliased T-maze:

- - At the start, the observation reveals a binary cue.
- - Along the corridor, observations are aliased and do not reveal the cue.
- - At the junction, the correct action depends on the cue.
- - After the junction decision, the trial resets with a new cue.

The task is continuing in the sense that the agent keeps learning over many trials, but the accuracy metric is measured at trial-end decisions.

Methods

State constructions compared:

- - raw observation;
- - short history;
- - trace memory;
- - recurrent GVF features;
- - oracle cue memory.

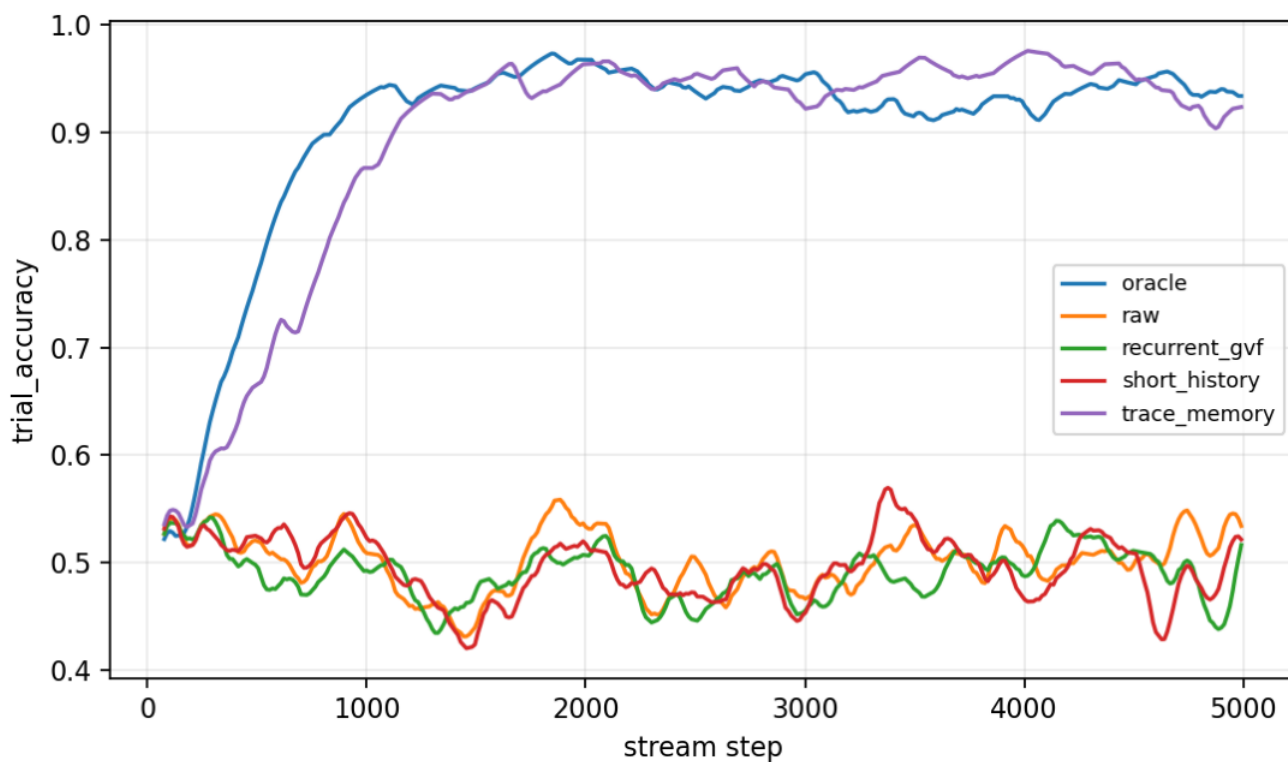
The oracle defines an upper bound. The trace memory baseline asks whether the task is solvable by simple non-deep memory. The GVF variant is the actual proposal mechanism.

Experimental Design

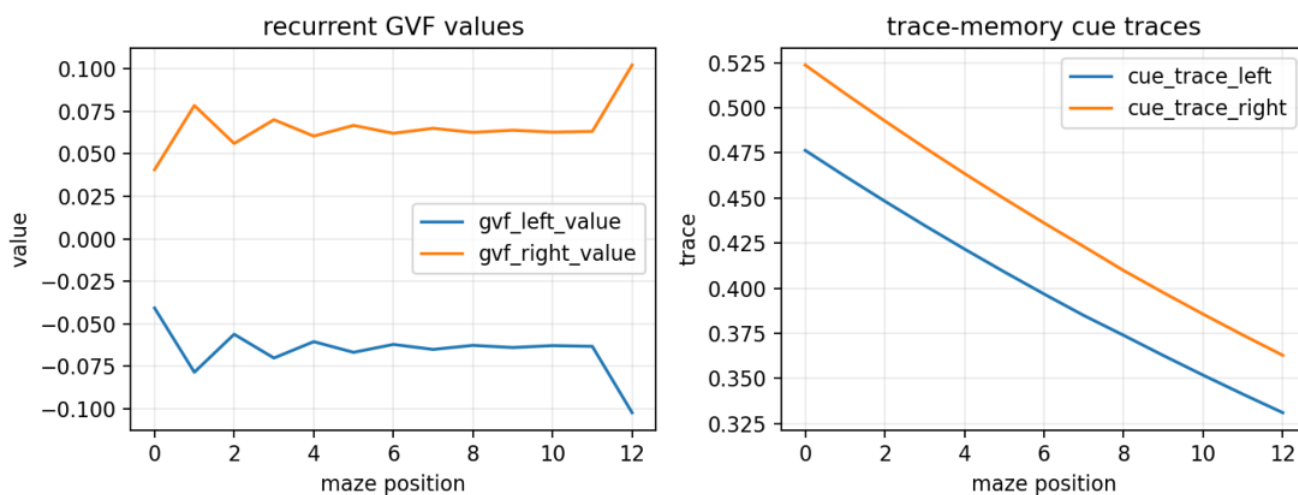
Current main pilot:

- - Maze length: 12.
- - Seeds: 0-4.
- - Steps: 5000.
- - Metrics: trial-end accuracy, average reward, GVF TD error, cue traces, position-level trajectories.
- - Result path:
experiments/alberta_core_rl/results/useful_gvfs_state/20260708T155740Z_main.

Figures:



Trial accuracy by state construction.



GVF values and cue traces by maze position.

Results

Trace memory and oracle memory solve the task, with seed-aware tail trial accuracy about 0.944 +/- 0.018 and 0.939 +/- 0.027. This establishes that the task is learnable with small non-deep state augmentation.

Raw observation, short history, and recurrent GVF remain near chance: about 0.512, 0.492, and 0.502. The current GVF state therefore does not carry the cue information needed by the control decision.

The trajectory figure supports this interpretation. The learned GVF signal does not show a clean cue-dependent trace that persists to the junction in the way the hand-coded trace does.

Analysis

This is not a failed experiment; it is a useful negative result. It separates three possibilities:

- - The task might be impossible under the course constraints.
- - The control algorithm might be broken.
- - The current GVF question/state design might be insufficient.

The trace and oracle baselines rule out the first possibility. The remaining conclusion is that the GVF feature, as designed, is not the right predictive state. That is exactly the kind of boundary condition a serious GVF proposal should expose.

Threats To Validity

The current GVF design is only one design. It should not be used to claim that GVFs cannot construct state.

The report currently lacks a direct linear probe of cue information from the GVF feature. The conclusion is inferred from control accuracy and trajectory plots.

The maze has a fixed length. A richer result should sweep corridor length and cue semantics.

The GVF cumulant/discount choices may not be aligned with the hidden variable. A fairer positive test requires question-design diagnostics before control evaluation.

Reviewer Critique And Revisions

GVF reviewer:

- - A GVF that predicts an easy but irrelevant signal is not useful state.

Revision made:

- - Added trace memory and oracle memory, plus position-level trajectory plots.

Strict baseline reviewer:

- - If trace memory solves the task, the GVF must approach or beat that baseline to be promoted as a positive state-construction result.

Revision required:

- - Add cue-decodability probes, hidden-cue correlation plots, and redesigned

cumulants tied to future cue-relevant events.

Conclusion

The current GVF Predictive State proposal is an independent negative result. It shows that the task is solvable by cheap memory but not by the current learned GVF feature. The next iteration should not merely tune hyperparameters; it should redesign the GVF questions so that the learned prediction is demonstrably tied to the hidden cue and useful for the junction decision.

Reproduction

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
--config experiments/alberta_core_rl/configs/useful_gvfs_state/config_main.json
```